1 Intrinsic dyadic multiplicative scale

Source: `PrimeTensor/Scale.lean`

What this file establishes

`PrimeTensor/Order.lean` left closeness parameterised by an arbitrary tolerance δ, admissible when $\delta > 1$. This file replaces that parameter by a single fixed ruler — the magnitude 2 — and recovers arbitrarily fine tolerances not by shrinking the ruler but by *squaring the ratio being measured*. Closeness at level ℓ says that the oriented ratio, squared $\ell - 1$ times, is still below 2.

The substitution is exact and it is forced. Refining a dyadic tolerance directly would mean comparing against $2^{1/2}, 2^{1/4}, 2^{1/8}, \dots$, and none of those is a positive rational, so none of them is an element of MulRat: the tolerances one wants are not expressible in the object language. The equivalent condition on the squared ratio is, and it uses only multiplication and the order already built. This is what the module header means by admitting no rational square root.

1.1 The ruler

Definition 1.1 (`twoRatio` and `MulRat.two`)**.**

$$\begin{aligned} \mathrm{twoRatio} &\equiv \langle\, \mathrm{ofBag}(\mathsf{factor}\ \mathrm{primeTwo}\ \mathsf{one}),\ 1 \,\rangle && : \mathrm{PrimeRatio}, \\ \mathrm{two} &\equiv \mathrm{ofRatio}(\mathrm{twoRatio}) && : \mathrm{MulRat}. \end{aligned}$$

```
def twoRatio : PrimeRatio :=
  ⟨PrimeMultiset.ofBag (.factor primeTwo .one), 1⟩

/-- Distinguished multiplicative ruler. -/
def two : MulRat :=
  ofRatio twoRatio
```

The numerator is the one-atom barcode carrying the prime 2 and nothing else, and the denominator is the pivot, so the pair denotes 2/1. Note that the ruler is built, not postulated: it is an ordinary element of the carrier, assembled from primeTwo of `PrimeTensor/Prime.lean`.

Lemma 1.2 (`MulRat.one_lt_two`)**.** $1 < \mathrm{two}$ *in* MulRat*.*

Proof. Both sides are classes of literals, so the lifted order reduces definitionally to the representative-level comparison $\mathrm{PrimeRatio.Lt}(1, \mathrm{twoRatio})$, which by `PrimeTensor/Ratio.lean` unfolds to

$$\mathrm{eval}(1.\mathrm{upper}) \cdot \mathrm{eval}(\mathrm{twoRatio.lower}) \ < \ \mathrm{eval}(\mathrm{twoRatio.upper}) \cdot \mathrm{eval}(1.\mathrm{lower}).$$

Simplification evaluates the four components: the unit contributes 1 on both sides by one_upper, one_lower and eval_one, while the numerator of the ruler evaluates through eval_ofBag and eval_factor to $\mathrm{primeTwo.value} \cdot 1$. The goal becomes $1 \cdot 1 < 2 \cdot 1$, closed by `norm_num` once primeTwo is unfolded. □

So the ruler is an admissible tolerance in the sense of `PrimeTensor/Order.lean`, and being fixed, it is admissible once and for all — no hypothesis $1 < \delta$ has to be carried anywhere below.

1.2 Repeated squaring

Definition 1.3 (`MulRat.scalePow`)**.** For r : MulRat, define scalePow$(r, -)$: Depth $\to$ MulRat by structural recursion on the level:

$$\mathrm{scalePow}(r, \mathsf{one}) = r, \qquad \mathrm{scalePow}(r, \mathsf{succ}\, d) = q \cdot q \quad \text{where } q = \mathrm{scalePow}(r, d).$$

```
def scalePow (r : MulRat) : Depth → MulRat
  | .one => r
  | .succ d =>
     let q := scalePow r d
     q * q
```

Proposition 1.4 (The effective exponent)**.** scalePow$(r, \ell) = r^{2^{|\ell|-1}}$*. In particular the exponents realised, as ℓ ranges over the depths, are* $1, 2, 4, 8, \ldots$ *— the powers of two, and only those.*

Proof. By induction on ℓ. For $\ell = \mathsf{one}$, $|\ell| = 1$ and the value is $r = r^{2^0}$. For $\ell = \mathsf{succ}\, d$, the induction hypothesis gives $q = r^{2^{|d|-1}}$, and $q \cdot q = r^{2 \cdot 2^{|d|-1}} = r^{2^{|d|}} = r^{2^{|\mathsf{succ}\, d|-1}}$. □

The definition squares the running value rather than multiplying by r, so the level counts *doublings of the exponent*, not the exponent itself. Level ℓ therefore reaches exponent $2^{|\ell|-1}$ in $|\ell|$ steps: the ladder climbs geometrically in a linear number of levels.

Lemma 1.5 (`MulRat.scalePow_one`)**.** scalePow$(1, \ell) = 1$ *for every level* ℓ*.*

```
@[simp] theorem scalePow_one : ∀ d : Depth, scalePow (1 : MulRat) d = 1
  | .one => rfl
  | .succ d => by
     change scalePow (1 : MulRat) d * scalePow (1 : MulRat) d = 1
     rw [scalePow_one d, one_mul]
```

Proof. By structural recursion on the level. At one the value is the argument, so the claim is rfl. At $\mathsf{succ}\, d$ the value is $q \cdot q$ with $q = \mathrm{scalePow}(1, d)$; the induction hypothesis rewrites q to 1, and one_mul from `PrimeTensor/MulRat.lean` finishes. Note the recursive call is made explicitly on d, the lemma being stated as a $\forall$ and proved by pattern matching rather than by induction. □

The unit is thus fixed by the whole ladder, which is exactly what reflexivity of the closeness relation will need.

1.3 Closeness at an intrinsic scale

Definition 1.6 (`MulRat.ScaleWithin`)**.** For a level ℓ and a, b : MulRat,

$$\mathrm{ScaleWithin}(\ell, a, b) \equiv (\mathrm{scalePow}(\mathrm{ratio}(a, b), \ell) < \mathrm{two}) \wedge (\mathrm{scalePow}(\mathrm{ratio}(b, a), \ell) < \mathrm{two}).$$

```
def ScaleWithin (level : Depth) (a b : MulRat) : Prop :=
  scalePow (ratio a b) level < two ∧
  scalePow (ratio b a) level < two
```

Proposition 1.7 (What the levels measure)**.** ScaleWithin(ℓ, a, b) *holds precisely when both oriented ratios lie strictly below* $2^{1/2^{|\ell|-1}}$*. Increasing the level therefore tightens the requirement, and the effective tolerances*

$$2,\ \sqrt{2},\ \sqrt[4]{2},\ \sqrt[8]{2},\ \ldots$$

decrease towards 1*.*

Proof. By Proposition 1.4 the first conjunct is $r^{2^{|\ell|-1}} < 2$ where $r = \mathrm{ratio}(a, b)$, and for $r > 0$ this is equivalent to $r < 2^{1/2^{|\ell|-1}}$, the map $x \mapsto x^n$ being strictly increasing on the positive reals for $n \geq 1$. The same applies to the second conjunct. □

Design note 1.8 (Why squaring the ratio and not shrinking the ruler). Proposition 1.7 makes the design constraint visible. The tolerances the ladder realises are $2^{1/2^k}$, and for $k \geq 1$ every one of them is irrational: $\sqrt{2}$ already is. So none of them is an element of MulRat, which by construction contains exactly the positive rationals. A definition of the form "$\mathrm{ratio}(a, b) < \delta_k$" with δ_k the k-th tolerance is therefore not expressible here at all — not merely inconvenient, but absent from the object language.

Squaring the ratio states the same condition using only what is available: a multiplication, the fixed ruler two, and the order of `PrimeTensor/Order.lean`. This is the sense in which the scale is *intrinsic*: the refinement is carried by the depth index, a piece of the object language, rather than by a sequence of tolerances that would have to be imported from outside it.

The same reading explains the other two exclusions the header names. There is no additive midpoint because there is no addition; and there is no zeroth scale because levels are depths, so the coarsest tolerance is the ruler itself and the ladder has a bottom rung rather than a degenerate one.

Remark 1.9 (Level one is the old notion). $\mathrm{scalePow}(r, \texttt{one})$ is r by definition, so $\mathrm{ScaleWithin}(\texttt{one}, a, b)$ is literally $\mathrm{ratio}(a, b) < \mathrm{two} \wedge \mathrm{ratio}(b, a) < \mathrm{two}$, which is $\mathrm{Within}(\mathrm{two}, a, b)$ in the sense of `PrimeTensor/Order.lean` — the two unfold to the same proposition. The new relation is thus a refinement of the old one at a particular tolerance, not a replacement for it. The file does not state this.

Lemma 1.10 (`scaleWithin_refl` and `scaleWithin_symm`)**.** $\mathrm{ScaleWithin}(\ell, a, a)$ *holds for every level ℓ and every a; and* $\mathrm{ScaleWithin}(\ell, a, b)$ *implies* $\mathrm{ScaleWithin}(\ell, b, a)$.

Proof. For reflexivity, $\mathrm{ratio}(a, a) = 1$ by ratio_self of `PrimeTensor/Order.lean`, then Lemma 1.5 collapses the whole ladder to 1, and each conjunct becomes Lemma 1.2. Symmetry is the exchange of the two conjuncts. □

Remark 1.11 (Reflexivity is now unconditional). Compare within_refl in `PrimeTensor/Order.lean`, which required the hypothesis $1 < \delta$ and could not do without it. Here there is no hypothesis at all: the tolerance is two, fixed, and Lemma 1.2 discharges the admissibility condition once inside this file. Every later statement about ScaleWithin is free of it. This is a small illustration of the general point — fixing the ruler converts a carried hypothesis into a proved lemma.

Remark 1.12 (Monotonicity in the level is not proved). The levels ought to be nested: closeness at a finer level should imply closeness at a coarser one, so that the relations form a decreasing family. They do — if $r^{2^{k-1}} \geq 2$ then squaring gives $r^{2^k} \geq 4 > 2$, so contrapositively $\mathrm{ScaleWithin}(\texttt{succ}\, d, a, b)$ implies $\mathrm{ScaleWithin}(d, a, b)$ — but the file states no such lemma, and the argument as given needs a comparison fact about MulRat beyond the order lemmas available so far. Nor is there any composition rule across two levels, the analogue of a triangle inequality, which `PrimeTensor/Order.lean` also lacked. Both are what a Cauchy notion built on this scale will require.

Summary of the correspondence

Lean declaration	Kind	Here
`MulRat.twoRatio`	definition	Def. 1.1
`MulRat.two`	definition	Def. 1.1
`MulRat.one_lt_two`	theorem	Lem. 1.2
`MulRat.scalePow`	definition	Def. 1.3
`MulRat.scalePow_one`	simp thm	Lem. 1.5
`MulRat.ScaleWithin`	definition	Def. 1.6
`MulRat.scaleWithin_refl`	theorem	Lem. 1.10
`MulRat.scaleWithin_symm`	theorem	Lem. 1.10

Every declaration of `PrimeTensor/Scale.lean` appears above. The file contains no `sorry`, no axiom, and no unproved named proposition. Propositions 1.4 and 1.7, and Remarks 1.9 and 1.12, are stated for the reader and are not declarations of the file; the first two are the arithmetic reading of the ladder, and they are what Design note 1.8 rests on.